
MSC-CMA-ES: STRUCTURE-AWARE RESTARTS FOR CMA-ES VIA CYCLIC NEAREST-BETTER BASIN DISCOVERY

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ABSTRACT

CMA-ES is, per run, a local optimizer; multimodal search relies on restart strategies such as IPOP and BIPOP, which draw every restart uniformly and reuse no information from previous evaluations. Multi-Start Clustering CMA-ES (MSC-CMA-ES) makes restarts structure-aware: in alternating cycles, a Sobol pre-sample is partitioned into approximate basins of attraction by nearest-better clustering, restarts are seeded basin by basin with locally scaled step sizes and population sizes, redundant basin visits are detected and excluded, and the remaining budget is spent on a budget-bounded, tolerance-disabled local refinement of the best-so-far solution. We evaluate the method on four CEC 2014, 2017, 2020, 2022 suites at their official budgets, across ten (suite, dimension) cells with dimensions 5–30, 51 runs per function, against BIPOP-CMA-ES and five differential-evolution algorithms (ARRDE, jSO, j2020, NLSHADE-RSP, LSRTDE). Read per function class, MSC-CMA-ES leads on one class, is mixed on a second, and trails on the third. On composition functions, MSC-CMA-ES attains the best value on all four aggregate measures, with $2.7\times$ the fixed-budget target coverage of BIPOP-CMA-ES—the highest composition coverage of any algorithm evaluated. On basic functions, it achieves the best (lowest) median error but exhibits a lower deep-target coverage—the measured price of spending budget on landscape discovery. On hybrid functions, both CMA variants trail the leading DE algorithms; in these data the deficit is shared by the two CMA variants. All results and scripts are publicly available.

Keywords CMA-ES, BIPOP, basin of attraction, nearest-better clustering, restart strategy, budget scaling

1 Introduction

The Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [7] adapts a single multivariate Gaussian and is therefore, per run, a local optimizer: on a multimodal landscape it converges to one basin of attraction. Multi-basin search is traditionally delegated to restart strategies, with IPOP [2] and BIPOP [8] serving as the de facto standards. Even two decades after its introduction, BIPOP-CMA-ES remains one of the most robust general-purpose baselines for continuous black-box optimization. Both strategies, however, leave information unused: every restart draws x_0 uniformly and sets $\sigma_0 = (u - l)/4$, so the topographic information already paid for by previous evaluations never informs the next restart. On multi-basin landscapes this has two recurring costs: evaluations are spent contracting

a domain-scale σ_0 onto narrow local structure, and successive restarts re-enter basins that earlier runs have already exhausted. Composition-type benchmark functions are constructed by combining rotated and shifted base functions around a randomly located global optimum and several deep, randomly located local optima with differing local properties [12, 3, 25, 23]; a uniformly initialized restart has no mechanism to prefer the global basin over the deceptive ones.

MSC-CMA-ES makes restarts structure-aware. The search proceeds in cycles. Each cycle opens with Phase 0. In this phase, a Sobol design of $M = 2^n$ points is evaluated. These points are then partitioned into approximate basins of attraction by Nearest-Better Clustering [16, 17]. The cutting threshold is selected automatically by a staircase scan of the nearest-better edge-length distribution.

Phase 1 seeds one CMA-ES restart per basin, moving from the smallest to the largest. Each restart initializes from the basin’s best sampled point. The initial step size σ_0 is computed from the coordinate-wise spread of the basin around its elite centre. The initial population size is proportional to the basin’s sample size. A k -NN membership vote identifies the basin into which each restart converges. Any basin resolved twice is excluded from further restarts within that cycle.

Successive cycles alternate between two complementary configurations of this pipeline. The first configuration partitions the sample into many small basins. These are probed by short, small-population runs. The second configuration partitions the space into few large basins. These are searched by long, large-population runs.

Sobol sequences are naturally nested. Therefore, every second cycle simply re-clusters a prefix of the previous cycle’s sample. This design spends no additional evaluations on Phase 0.

When the remaining budget no longer supports a full cycle, a final refinement stage begins. This stage spends all remaining evaluations on a single CMA-ES run from the incumbent solution. All internal convergence criteria are disabled for this run. Consequently, the remaining budget—and not a tolerance floor—terminates the optimization. The full algorithm is specified in Section 2.

In Section 3 we report the official-budget results of seven algorithms—MSC-CMA-ES, BIPOP-CMA-ES, and five differential-evolution baselines (ARRDE [15], jSO [5], j2020 [6], NLSHADE-RSP [19], and LSRTDE [20], [21])—read along two cross-sections:

1. **By function class** (Table 3): on the composition class MSC-CMA-ES attains the lowest summed mean, median, and best error, and the highest target coverage— $2.7\times$ that of its BIPOP-CMA-ES parent.
2. **By dimension** (Table 2): the lowest summed mean and median at every $D \leq 20$, the margin over the nearest baseline ranging from a few percent at $D=10$ to roughly ten percent at $D=20$.

Section 3 also reports where the advantage does not hold: a lower target coverage on the basic class, a several-fold deficit to the DE baselines on the hybrid class, and a spread within roughly a factor of two across all seven algorithms at $D=30$.

The complete source code for MSC-CMA-ES, implemented in Python and optimized for the Intel Distribution for Python, is provided in the Supplementary Material¹ alongside the execution scripts. This material also contains all computational results.

Contributions

1. A **staircase ϕ selection procedure** for NBC that determines the cutting threshold automatically from the edge-length distribution of the nearest-better tree, eliminating manual ϕ tuning.
2. **Per-basin step-size and population-size adaptation**, replacing the fixed BIPOP heuristic $\sigma_0 = (u - l)/4$ with a σ_0 computed from the discovered basin geometry and a CMA population size proportional to the basin’s sample size.
3. A **convergence-tracking mechanism** (k -NN majority vote) that detects redundant basin visits and excludes twice-resolved basins from further restarts within the cycle.
4. An **alternating two-configuration cycle schedule** that re-clusters the same landscape at two granularities—many small basins probed by short runs, few large basins searched by long runs—with **zero-cost Phase-0 reuse** of the sample across cycles via the nested-prefix property of Sobol sequences.

¹https://github.com/snenovgmailcom/cma_es_project

Algorithm 1 MSC-CMA-ES (skeleton; full version in Algorithm 3)**Require:** objective f , box Ω , budget T , configuration pair (θ_C, θ_B)

```

1: reserve refinement budget:  $T_{\text{main}} \leftarrow T - \lfloor r_C T \rfloor$ 
2: for cycle  $c = 0, 1, 2, \dots$  while the budget gate (Sec. 2.5) holds do
3:    $\theta \leftarrow \theta_C$  if  $c$  even else  $\theta_B$  ▷ alternating configurations
4:   Phase 0: sample  $\Omega$ , cluster into basins  $B_1 \leq \dots \leq B_K$  ▷ Sec. 2.2; odd cycles reuse the previous sample
5:   Phase 1: one CMA-ES restart per basin, smallest first ▷ Sec. 2.4
6: end for
7: if at least  $10D$  evaluations remain then
8:   Refinement: single CMA-ES from  $x^{\text{best}}$  on the remaining budget ▷ Sec. 2.6
9: end if
10: return incumbent  $x^{\text{best}}$ 

```

2 The algorithm

2.1 Setting and notation

We minimize a continuous function $f : \Omega \rightarrow \mathbb{R}$ on a box $\Omega = \prod_{j=1}^D [l_j, u_j] \subset \mathbb{R}^D$ under a fixed budget of T function evaluations. Distances are measured in normalized coordinates, $\hat{x}_j = (x_j - l_j)/(u_j - l_j)$, so that $\hat{\Omega} = [0, 1]^D$.

The method runs a sequence of *cycles* and closes with a single refinement. Each cycle has two phases: *Phase 0* (Section 2.2) evaluates a space-filling sample of Ω and partitions it into *basins* by nearest-better clustering; *Phase 1* (Section 2.4) then launches one CMA-ES restart per basin, in ascending order of basin size. Cycles repeat while a budget gate (Section 2.5) admits another one; once the main budget is spent, a final CMA-ES *refinement* (Section 2.6) consumes whatever evaluations remain, polishing the incumbent x^{best} .

Two fixed sub-configurations, θ_C and θ_B , drive the cycles in strict alternation: cycle 0 uses θ_C , cycle 1 uses θ_B , cycle 2 uses θ_C , and so on. Both are fixed across all dimensions and budgets (Table 1); we write $\theta = (\theta_C, \theta_B)$ for the pair, and detail their contrasting roles in Section 2.5. Algorithm 1 gives the skeleton; the fully specified version is Algorithm 3.

2.2 Phase 0: basin discovery

Design. Phase 0 evaluates a Sobol sample $P = \{x_1, \dots, x_M\} \subset \Omega$ with $M = 2^n$ and records $f_i = f(x_i)$, $i = 1, \dots, M$.

Nearest-better clustering. Every sampled point is connected to its nearest *better* neighbour, in normalized coordinates:

$$\text{nb}(i) = \min \left\{ \underset{j: f_j < f_i}{\text{argmin}} \|\hat{x}_j - \hat{x}_i\| \right\}, \quad \ell_i = \|\hat{x}_{\text{nb}(i)} - \hat{x}_i\|. \quad (1)$$

The sample-best point has no better neighbour and becomes the root; the result is a tree on the M sample points with $M - 1$ edges. Relation (1) is evaluated exactly: the nearest better neighbour is taken over all better points, with no k -NN cap and no fallback edges, at a cost of $O(M^2 D)$. We follow the two heuristic rules of Preuss [16, 17] to build the NBC partition on top of this tree.

Rule 1 (long-edge cut). Let

$$\bar{\ell} = \frac{1}{|\{i : \text{nb}(i) \geq 0\}|} \sum_{i: \text{nb}(i) \geq 0} \ell_i$$

be the mean edge length. Cut all edges exceeding a multiple of the mean:

$$\text{cut edge } (i, \text{nb}(i)) \quad \text{if} \quad \ell_i > \phi \bar{\ell}, \quad (2)$$

where $\phi > 0$ is the cutting threshold. Point i whose edge is cut becomes a new basin root.

Rule 2 (hub detection). A point j with high indegree whose own outgoing edge is much longer than its incoming edges is a local attractor incorrectly connected to a distant parent. Let $\text{in}(j) = \{i : \text{nb}(i) = j\}$ in the graph pruned by Rule 1, and $\bar{\ell}_{\text{in}}(j) = |\text{in}(j)|^{-1} \sum_{i \in \text{in}(j)} \ell_i$. Then:

$$\text{cut edge } (j, \text{nb}(j)) \quad \text{if} \quad |\text{in}(j)| \geq n_{\min} \quad \text{and} \quad \frac{\ell_j}{\bar{\ell}_{\text{in}}(j)} > b, \quad (3)$$

where $n_{\min}$ is the minimum indegree and $b > 0$ is the ratio threshold, a component of the tuned configuration (Table 1). After both rules are applied, basin labels are obtained by pointer jumping on the pruned parent array. Let $\mathcal{T} : \{1, \dots, M\} \rightarrow \{1, \dots, M\}$ be the pruned-parent map, with $\mathcal{T}(i) = \text{nb}(i)$ on the edges retained after Rules 1–2 and $\mathcal{T}(i) = i$ on the roots (points whose edge was cut or which are themselves the NB-tree root). Pointer jumping iterates $\mathcal{T}^{(t+1)} = \mathcal{T} \circ \mathcal{T}^{(t)}$; since the pruned forest has depth $O(\log M)$, the fixed point

$$\text{root}(i) = \lim_{t \rightarrow \infty} \mathcal{T}^{(t)}(i)$$

is reached in $O(\log M)$ iterations. The resulting partition is $\mathcal{S} = \mathcal{C}_1 \sqcup \dots \sqcup \mathcal{C}_B$, where basins smaller than the minimum size $s_{\min}$ (Table 1) are discarded as noise.

2.3 Staircase ϕ selection

The threshold ϕ is the main free parameter of NBC [16], and its choice is known to be landscape-dependent. Rather than using a fixed ϕ , we automatically select it to achieve a target number of basins n_{target} .

As ϕ decreases from $+\infty$ to 0, the Rule-1 cut set changes only when ϕ crosses a ratio $\ell_i/\bar{\ell}$. Sort these ratios in non-increasing order to obtain the candidate ladder

$$\phi_k^* = \ell_{(k)}/\bar{\ell}, \quad \phi_1^* \geq \phi_2^* \geq \dots \geq \phi_{M-1}^*, \quad (4)$$

where $\ell_{(1)} \geq \dots \geq \ell_{(M-1)}$ are the sorted NB-tree edge lengths and $\bar{\ell}$ their mean.

Lemma 1. *Assume the edge lengths are pairwise distinct. Let $n_b(\phi)$ be the number of components after Rule 1. Then n_b is non-increasing in ϕ , piecewise constant with jumps exactly at the ϕ_k^* , and $n_b(\phi) = k + 1$ on the open interval (ϕ_{k+1}^*, ϕ_k^*) .*

Proof. For $\phi \in (\phi_{k+1}^*, \phi_k^*)$ the cut set $\{i : \ell_i > \phi\bar{\ell}\}$ consists exactly of the edges of rank $1, \dots, k$. The NB graph is a tree on M vertices with $M - 1$ edges, so deleting k of them leaves exactly $k + 1$ connected components. Monotonicity and the jump locations follow since the cut set grows by one edge each time ϕ crosses a ϕ_k^* from above. With ties, consecutive intervals merge and the statement holds with multiplicities. $\square$

Algorithm. Let $\phi_1^* \geq \phi_2^* \geq \dots \geq \phi_{M-1}^*$ denote the non-root edge ratios $\ell_i/\bar{\ell}$ sorted in non-increasing order (the candidate ladder of Eq. 2), and define

$$\phi^{(k)} = \frac{1}{2}(\phi_k^* + \phi_{k+1}^*), \quad k = 1, \dots, M - 2.$$

At each $\phi^{(k)}$ we apply Rule 1 and Rule 2 to the cached NB-tree, pointer-jump the pruned parent array to labels, and record the count of useful basins

$$n_{\text{useful}}(\phi^{(k)}) = |\{a : |\mathcal{C}_a(\phi^{(k)})| \geq s_{\min}\}|,$$

where $\mathcal{C}_a(\phi)$ are the components induced at threshold ϕ ; this scan incurs no objective evaluations. The selected threshold is

$$k^* = \min\{k \in \{1, \dots, M - 2\} : n_{\text{useful}}(\phi^{(k)}) \geq n_{\text{target}}\}, \quad \phi_{\text{used}} = \phi^{(k^*)},$$

and when this set is empty we fall back to $\phi_{\text{used}} = \frac{1}{2}\phi_{M-1}^*$, the finest Rule-1 pruning (every non-root edge cut); basins below $s_{\min}$ are then discarded as usual.

By Lemma 1 the raw Rule-1 component count is monotone in ϕ with unit jumps exactly at the ϕ_k^* , so the ladder enumerates every distinct Rule-1 pruning. The composed map $\phi \mapsto n_{\text{useful}}(\phi)$, however, is *not* monotone: the $s_{\min}$ filter removes components as they shrink below $s_{\min}$ when ϕ decreases, and Rule 2 cuts on the already-pruned indegrees. A bisection over ϕ is therefore unsound, while a single descending pass over the ladder is both sufficient and necessary, and $\phi^{(k^*)}$ is by construction the largest threshold attaining the target.

Let us mark that the proposed procedure applies Rule 1 and Rule 2 cuts to the stored raw NB-tree without any new function evaluations: it operates purely on the cached parent array and edge lengths from the initial NBC construction. The staircase may call this routine hundreds of times during the ϕ scan, but each call is $O(M \log M)$ (vectorized pointer jumping), making the total cost negligible compared to the M function evaluations.

2.4 Phase 1: structure-aware restarts

Basins are processed in ascending order of size. For each basin B not yet excluded (see below), one CMA-ES [7] restart is launched with three quantities derived from the basin geometry:

Algorithm 2 Staircase ϕ selection**Require:** NB-tree parent array, edge lengths, mean $\bar{\ell}$, target n_{target} , min basin size s_{min} **Ensure:** ϕ_{used} , history $[(\phi, n)]$

```

1: Sort ratios  $\ell_i/\bar{\ell}$  descending:  $\phi_1^* \geq \phi_2^* \geq \dots \geq \phi_{M-1}^*$ 
2: for  $k = 1, 2, \dots, M-1$  do
3:    $\phi \leftarrow (\phi_k^* + \phi_{k+1}^*)/2$ 
4:   Apply Rule 1 + Rule 2 at  $\phi$  ▷ no new evals
5:   Count basins with  $|\mathcal{C}_a| \geq s_{\text{min}}$ 
6:   Record  $(\phi, n_{\text{useful}})$ 
7:   if  $n_{\text{useful}} \geq n_{\text{target}}$  then
8:     return  $\phi$ 
9:   end if
10: end for
11: return  $\phi_{M-1}^*/2$  ▷ fallback if target unreachable

```

Initial point. $x_0 = \operatorname{argmin}_{x \in B} f(x)$, the best sampled point of the basin.**Initial step size.** Let c be the *elite centre*, the mean of the best $\max\{1, \lfloor \varepsilon |B| \rfloor\}$ points of B by f -value, and let

$$s_j = Q_{75}(|x_{ij} - c_j| : x_i \in B) \quad (j = 1, \dots, D) \quad (5)$$

be the per-axis upper-quartile spread around it. Then

$$\sigma_0 = \max\left(\frac{\operatorname{median}_j s_j}{\delta(D)}, 1\right), \quad \delta(D) = \delta_{\text{ref}} \sqrt{10/D}, \quad (6)$$

where δ_{ref} is the configuration's divisor anchored at $D = 10$ and the $\sqrt{10/D}$ factor is the standard CMA box-scaling law; the lower bound 1 is a safety net against degenerate basins.**Population size.**

$$\lambda = \operatorname{clamp}(\lceil \rho |B| \rceil, \lambda_H, \lambda_{\text{max}}), \quad \lambda_H = 4 + \lfloor 3 \ln D \rfloor, \quad (7)$$

i.e. proportional to the basin's sample size between the default λ_H and a configuration cap λ_{max} .**Stopping.** A restart stops at the first of: (i) an internal CMA-ES criterion with tolerances $\text{tol}_{\text{fun}} = 10^{-\tau_f}$, $\text{tol}_x = 10^{-\tau_x}$; (ii) absolute fitness convergence of the current population, $\max(F) - \min(F) < s_{\text{tol}}$; (iii) budget exhaustion.**Convergence tracking and exclusion.** Basin membership of any point x is decided by a k -NN majority vote ($k = 5$) over the labelled Phase-0 sample. Before a restart is launched, its x_0 is probed and the restart is skipped if the vote maps it into an excluded basin. After a restart finishes, the vote of its best point identifies the basin it converged into; a basin that has received two convergences is excluded from all further restarts of the current cycle. Exclusion is reset at the start of each cycle: the next cycle re-clusters the landscape at a different granularity, so basin identities do not carry over.

2.5 Cycles, alternating configurations, and sample reuse

The pair (Phase 0, Phase 1) constitutes one *cycle*. Cycles repeat with alternating configurations, $\theta_C, \theta_B, \theta_C, \dots$: the C configuration partitions the sample into many small basins probed by short, small-population runs ($n_b = 25$, $s_{\text{min}} = 7$, $\lambda_{\text{max}} = 12$), the B configuration into few large basins searched by long, large-population runs ($n_b = 5$, $s_{\text{min}} = 155$, $\lambda_{\text{max}} = 236$); all values in Table 1.

Since both configurations draw the same Sobol design size M , every second (odd) cycle reuses the points and f -values cached from the previous cycle. We re-cluster them under the odd cycle's own configuration (in particular its own b , s_{min} , and staircase target), spending *zero* additional evaluations on Phase 0. A new cycle is started only if the remaining main budget is at least $50D$ when the cycle is reuse-eligible, and $M + 50D$ otherwise.

2.6 Final refinement

A fraction r of the total budget is reserved at the start ($T_{\text{main}} = T - \lfloor rT \rfloor$); the reservation is a floor, not an allocation, since the refinement stage spends everything that remains. When the cycle loop terminates and at least $10D$ evaluations

Algorithm 3 MSC-CMA-ES

Require: f , box Ω , budget T , schedule (θ_C, θ_B)

```

1:  $T_{\text{main}} \leftarrow T - \lfloor r_C T \rfloor$ ;  $t \leftarrow 0$ ; cycle  $c \leftarrow 0$ 
2: while  $t < T_{\text{main}}$  and budget gate (Sec. 2.5) holds do
3:    $\theta \leftarrow \theta_C$  if  $c$  even else  $\theta_B$ 
4:   if  $c$  odd and previous sample reusable then
5:     reuse Sobol prefix; re-cluster with  $\theta$  ▷ 0 evaluations
6:   else
7:     evaluate Sobol design  $P$ ,  $|P| = M$  ▷  $t \leftarrow t + M$ 
8:   end if
9:   build NB tree (1); select  $\varphi$  by staircase; basins  $B_1 \leq \dots \leq B_K$ 
10:   $E \leftarrow \emptyset$  ▷ excluded basins, per cycle
11:  for  $B$  in ascending size, while  $t < T_{\text{main}}$  do
12:     $x_0 \leftarrow$  best point of  $B$ ; if  $\text{vote}(x_0) \in E$  skip
13:    run CMA-ES( $x_0, \sigma_0$  (6),  $\lambda$  (7)) until stop (Sec. 2.4)
14:     $\beta \leftarrow$  vote(best point of run); exclude  $\beta$  into  $E$  on its 2nd convergence
15:  end for
16:   $c \leftarrow c + 1$ 
17: end while
18: if  $T - t \geq 10D$  then
19:   run CMA-ES from  $x^{\text{best}}$  with all tolerance stops disabled until  $t = T$  ▷ refinement
20: end if
21: return  $x^{\text{best}}$ 
```

Table 1: The two configurations (Optuna [1] best trials, tuned on CEC2017 $D=10$ at the official budget; exact values in the repository). δ_{ref} is anchored at $D=10$ and rescaled by $\sqrt{10/D}$, Eq. (6); all other parameters are universal across dimensions and budgets.

Parameter	C (exploration)	B (exploitation)	Role
M (Phase-0 sample)	4096	4096	Sobol design size
n_b (target basins)	25	5	staircase target
$s_{\min}$ (useful basin)	7	155	min. basin size
b (Rule-2 ratio)	2.498	3.029	hub cut, Eq. (3)
m (Rule-2 in-degree)	3	3	hub cut
ε (elite frac.)	0.411	0.065	elite centre, Eq. (6)
δ_{ref}	5.989	1.661	σ_0 divisor
ρ (popsize frac.)	0.129	0.434	Eq. (7)
$\lambda_{\max}$	12	236	popsize cap
τ_f / τ_x	4 / 8	4 / 8	CMA tolerances
s_{tol}	13.82	13.82	population-range stop
r (refine reservation)	0.04	0.04	Sec. 2.6
k -NN vote	5	5	membership

remain, a single CMA-ES run is started from the x^{best} with $\sigma_0 = \min(s^*, \text{median}_j(u_j - l_j)/100)$, where s^* is the final step size of the restart that produced x^{best} (fallback: $s^* = \text{median}_j(u_j - l_j)/\delta(D)$), $\lambda = \max(\lambda_H, 10)$, and *all* tolerance-based stopping criteria disabled ($\text{tolfun} = \text{tolx} = \text{tolfunhist} = 0$, iteration and stagnation caps lifted). The run therefore terminates on the budget or on floating-point no-effect conditions only: the budget, not a tolerance, ends the optimization.

3 Experiments and Results

3.1 Protocol

We evaluate all algorithms on four benchmark suites from the IEEE Congress on Evolutionary Computation (CEC), observing strictly their official maximum evaluation budgets. Within the design envelope of the proposed method ($D \leq 20$) the experimental cells are: CEC2014 $D=10$ (10^5); CEC2017 $D=10$ (10^5); CEC2020 $D=5$ (5×10^4), $D=10$ (10^6), $D=15$ (3×10^6), and $D=20$ (10^7); and CEC2022 $D=10$ (2×10^5) and $D=20$ (10^6)—eight cells in all.

Two further cells outside the envelope, CEC2014 and CEC2017 at $D=30$ (3×10^5), are examined separately to probe the scalability boundary. Following the suite authors’ recommendation, the deprecated function $f2$ of CEC2017 is excluded for all algorithms, leaving 123 functions across the eight envelope cells.

The BIPOP-CMA-ES baseline is evaluated using the pycma reference implementation [9] with its native restart logic and the standard initial step size $\sigma_0 = (u - l)/4$. The five differential evolution (DE) baselines (ARRDE, jSO, j2020, NLSHADE-RSP, and LSRTDE) are executed via their respective minionpy C++ reference implementations [14], configured with the author-recommended default parameters.

MSC-CMA-ES operates under the alternating C/B schedule described in Section 2.5, utilizing the two fixed configurations detailed in Table 1. To assess the generalization capability and structural robustness of the approach, no per-suite or per-dimension hyperparameter tuning was conducted. Both configurations were tuned exactly once on the CEC2017 $D = 10$ cell at its official budget; consequently, all other parameters are universal across dimensions and budgets.

Each (algorithm, function, cell) configuration is evaluated over 51 independent runs using distinct random seeds from $\{0, \dots, 50\}$. Final optimization errors satisfying $|f(\mathbf{x}) - f^*| \leq 10^{-8}$ are floored to zero, matching the global target threshold floor. For each individual function, the Supplementary Material provides the mean, median, and best final errors, alongside the fixed-budget target coverage (FBTC): with 51 log-uniformly spaced targets $\tau \in [10^{-8}, 10^2]$, FBTC is the fraction of (target, run) pairs whose final error satisfies $e \leq \tau$. It lies in $[0, 1]$ per function and equals the terminal cross-section of the COCO-style target-hit profile; the result tables report its sum over the functions of each class or dimension. Statistical significance is evaluated at a per-function level via the Wilcoxon signed-rank test ($\alpha = 0.05$), applying the Benjamini–Hochberg false discovery rate (FDR) correction to within each (suite, dimension) cell, separately for each baseline–MSC-CMA-ES comparison, over the functions of that cell

Environment: Python 3.13.5 (anaconda3 env intelpython), NumPy 2.3.1, SciPy 1.15.3, pycma 4.4.2, minionpy 1.5.0. Hardware: Intel Xeon Platinum 8160 @ 2.10 GHz, 192 threads, 251 GiB RAM.

3.2 Per-Dimension Results at Official Budgets

Table 2 reports the summed mean, median, and fixed-budget target coverage (FBTC) per dimension at the official CEC budgets, for $D \in \{5, 10, 15, 20, 30\}$. At the official budget, MSC-CMA-ES attains the lowest summed mean and median error at every dimension up to and including $D=20$.

At $D=30$ (budget $10^4 D$) the lowest summed mean and median pass to the DE baselines, with MSC-CMA-ES mid-pack. This is the boundary of the design envelope of Section 3.1. Preuss notes that niching is not a universal cure: once a landscape is multimodal enough that not enough basins of attraction can be identified from a sample of affordable size, coordinated niching gains no clear advantage over uncoordinated restarts [16, 17]. The DE literature reaches the same boundary from the other side: ARRDE reports that restarts become less effective as dimension grows and de-emphasises them at large D [15]. A method built on NBC basin identification loses its low-dimensional advantage in this regime, which is consistent with the observed $D=30$ ordering; the present data do not by themselves isolate the cause. Two DE baselines (j2020, NLSHADE-RSP) diverge on a subset of the $D=30$ functions, raising their summed means more than an order of magnitude above the rest of the field.

On fixed-budget target coverage the picture is complementary: a DE baseline attains the highest FBTC at every dimension—NLSHADE-RSP at $D=5$ and $D=10$, j2020 at $D=15$, ARRDE at $D=20$, and LSRTDE at $D=30$ —while neither CMA variant leads coverage at any dimension. The summed errors live on very different scales across D (tens at $D=5$, thousands at $D=30$), so the per-dimension rows are the comparable unit; a cross-dimension total would be dominated by the $D=30$ divergences and is not reported.

3.3 Per-Type Results at Official Budgets

Table 3 reports the summed mean, median, best, and FBTC over the same official-budget cells, grouped by function class (basic, hybrid, composition); CEC2017 $f2$ is excluded, leaving 123 functions. The three classes give three different orderings.

On composition functions MSC-CMA-ES attains the best value on all four aggregates: the lowest summed mean, median, and best error, and the highest FBTC (9.69, against 3.56 for BIPOP-CMA-ES and 8.45 for the strongest DE baseline, NLSHADE-RSP). On basic functions it attains the lowest summed median (138.79) but the lowest FBTC of the field (25.46, against 27.04–30.63 for the others); its summed mean (221.10) is within 0.12 of the lowest (LSRTDE, 220.98). On hybrid functions the leading DE algorithms are several-fold to an order of magnitude ahead on mean and median (39.89/33.26 for jSO/LSRTDE against 265.11/294.35 for MSC-CMA-ES), and every DE baseline outcores

Table 2: Per-dimension totals at the official CEC budgets, $D \in \{5, 10, 15, 20, 30\}$ (function count in parentheses); 51 runs per function. Summed mean and median errors (lower better) and summed FBTC (higher better). Lowest error / highest FBTC per row in bold.

D	Metric		MSC-CMA	BIPOP-CMA	ARRDE	j2020	jSO	LSRTDE	NLSHADE-RSP
$D=5$ (10)	mean		74.37	391.23	159.08	189.14	452.12	482.34	227.22
	median		4.68	456.65	105.79	105.97	452.87	453.24	301.77
	FBTC		5.53	4.72	6.73	6.15	5.84	5.07	7.83
$D=10$ (81)	mean		4400.09	6266.20	4699.62	5408.53	5948.12	38389.88	5219.56
	median		4715.55	6322.05	4933.86	5336.90	6064.13	6388.68	5300.49
	FBTC		27.40	26.09	32.42	28.27	32.00	29.88	33.54
$D=15$ (10)	mean		280.88	638.44	407.84	557.14	906.86	938.01	563.23
	median		215.06	624.94	354.44	526.28	883.77	907.83	531.48
	FBTC		3.09	3.33	2.47	3.52	2.10	2.12	3.46
$D=20$ (22)	mean		1023.37	1366.02	1139.29	1747.65	1827.39	1811.99	1720.27
	median		1041.84	1246.21	1141.68	1775.53	1825.42	1798.42	1635.75
	FBTC		7.15	7.88	8.02	6.52	5.87	6.44	7.15
$D=30$ (59)	mean		15272.54	16831.29	13702.79	467983.24	11779.09	8630.72	438534.58
	median		14933.12	16162.84	13226.23	358024.96	11761.69	8395.07	277179.35
	FBTC		8.45	12.16	14.11	9.17	14.50	15.83	8.55

both CMA variants on FBTC; the two CMA variants sit together at the back, with the exception that one DE baseline, NLSHADE-RSP, is worse than both on mean and median.

These orderings are consistent with the per-dimension picture of Section 3.2. The composition class is the one the structure-aware restarts are designed to act on.

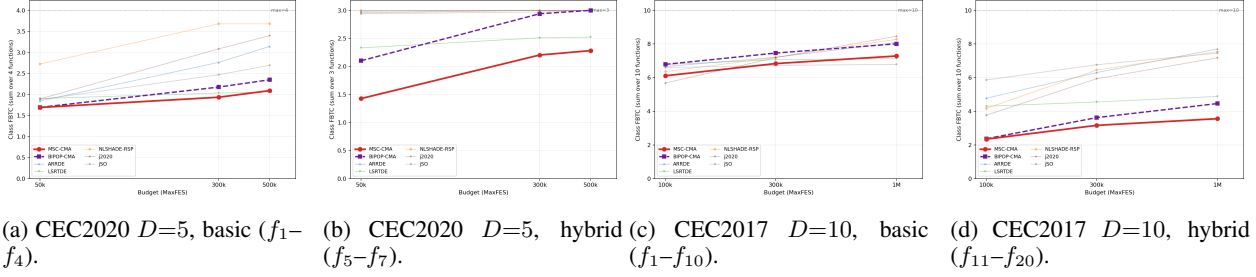
Table 3 is restricted to the cells with $D \leq 20$, the design envelope of Section 3.1: NBC basin identification is reliable only while basins remain recognizable from a sample of affordable size [16, 17]. The $D=30$ cells are reported per dimension in Table 2.

Table 3: Per-class totals over the eight official-budget cells with $D \in \{5, 10, 15, 20\}$; 51 runs per function; CEC2017 f2 (deprecated) excluded for all algorithms – 123 functions in total. Mean, median, and best are summed errors; FBTC is the summed fixed-budget target coverage. Lowest error or highest FBTC value per row in bold.

Class	Metric		MSC-CMA	BIPOP-CMA	ARRDE	j2020	jSO	LSRTDE	NLSHADE-RSP
Basic	mean		221.10	313.02	312.15	301.22	345.24	220.98	406.37
	median		138.79	231.64	249.80	251.47	337.87	194.11	437.91
	best		36.71	15.13	55.71	25.87	110.91	103.89	93.98
	FBTC		25.46	29.85	29.64	27.04	28.35	28.96	30.63
Hybrid	mean		265.11	357.75	63.74	254.48	39.89	58.17	524.30
	median		294.35	315.70	36.06	180.99	39.38	33.26	377.36
	best		12.09	6.54	1.33	27.70	3.22	2.83	9.28
	FBTC		8.01	8.60	14.47	12.25	15.20	12.40	12.90
Composition	mean		5292.49	7991.12	6029.94	7346.77	8749.36	41325.87	6799.60
	median		5543.97	8102.51	6249.91	7312.22	8848.94	9320.80	6954.21
	best		3008.56	4668.57	3211.60	4795.82	7962.90	7596.05	4037.85
	FBTC		9.69	3.56	5.52	5.17	2.26	2.15	8.45

3.4 Per-function comparison

At the per-function level the class pattern of Table 3 persists: MSC-CMA-ES is significantly stronger on composition functions, while the DE baselines are significantly better on the majority of hybrid functions. The complete per-function tables—mean, median, best, and FBTC with per-function Wilcoxon signed-rank tests (Benjamini–Hochberg FDR, $\alpha = 0.05$) against MSC-CMA-ES—are given in the Supplementary Material for every (suite, dimension) cell.



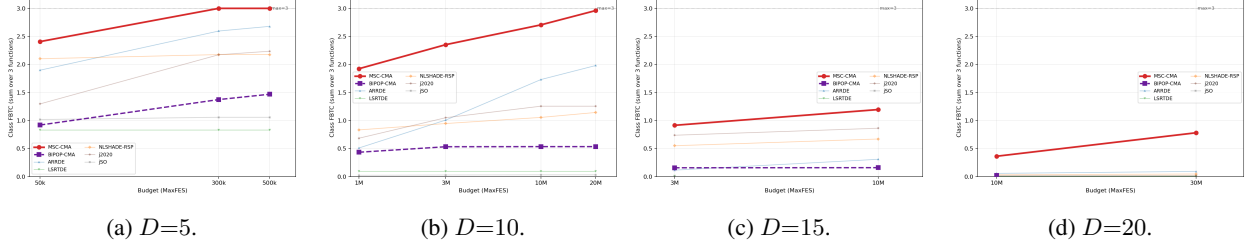


Figure 3: FBTC on the composition class (f_8 – f_{10} , class maximum 3) of CEC2020 at $D \in \{5, 10, 15, 20\}$.

while MSC-CMA-ES continues to increase its coverage monotonically over the evaluated budgets. On the $D = 20$ composition class it rises from approximately 1.1 at 2×10^6 evaluations to 2.6 (out of a maximum of 3) at 4×10^7 .

4 Discussion

Figures 4a and 4b track the seven algorithms on a composition class through four aggregate measures: worst-SUM, median-SUM, FBTC, and best-SUM. The three magnitude measures (worst, median, best) order the algorithms similarly; FBTC orders them differently.



Figure 4: Algorithm rankings on the composition classes across four aggregate measures: worst-SUM, median-SUM, FBTC, and best-SUM. The plots illustrate how normalized target coverage (FBTC) often disrupts the otherwise stable hierarchies established by raw error magnitude metrics.

On CEC2014 $D=10$ (Fig. 4a) the order is close to constant across worst-, median-, and best-SUM, while FBTC moves NLSHADE-RSP from third (median-SUM) to first and MSC-CMA-ES from first to third. On CEC2022 $D=20$ (Fig. 4b) FBTC moves BIPOP-CMA-ES from second (median-SUM) to fourth.

The two families aggregate different quantities. The SUM measures add raw errors and are dominated by the functions of largest scale, which on these classes are the unsolved ones; they order an algorithm by how far it reduces large-scale error, in its worst, median, and best run alike. FBTC averages per-function target coverage in $[0, 1]$, so each function contributes equally regardless of scale, and it orders an algorithm by how many targets it crosses. An algorithm that reaches the optimum on a few functions and misses the rest is high by FBTC and low by the SUM measures (NLSHADE-RSP); an algorithm that reduces the error on many hard functions without reaching their targets is the reverse (MSC-CMA-ES on CEC2014 $D=10$). The two are complementary: a composition-class comparison should report both—the SUM measures for error-scale reduction, FBTC for target coverage.

The same SUM/FBTC distinction explains the per-class profiles of Section 3.3. The basic-function profile—best median, worst coverage—is consistent with spending the opening budget on the Phase-0 design: the reconnaissance buys robustness across the bulk of runs (the median) at the cost of deep-target coverage (FBTC). On hybrid functions MSC-CMA-ES and BIPOP-CMA-ES track each other closely; the gap to the leading DE algorithms is therefore shared by both CMA variants rather than introduced by the restart mechanism, though the present data do not isolate its cause within the CMA family. Across suites the composition-class ordering is not fixed either: no single algorithm is universally optimal, and the advantage of MSC-CMA-ES on the CEC2020 composition class grows with dimension over the evaluated budgets (Section 3.5).

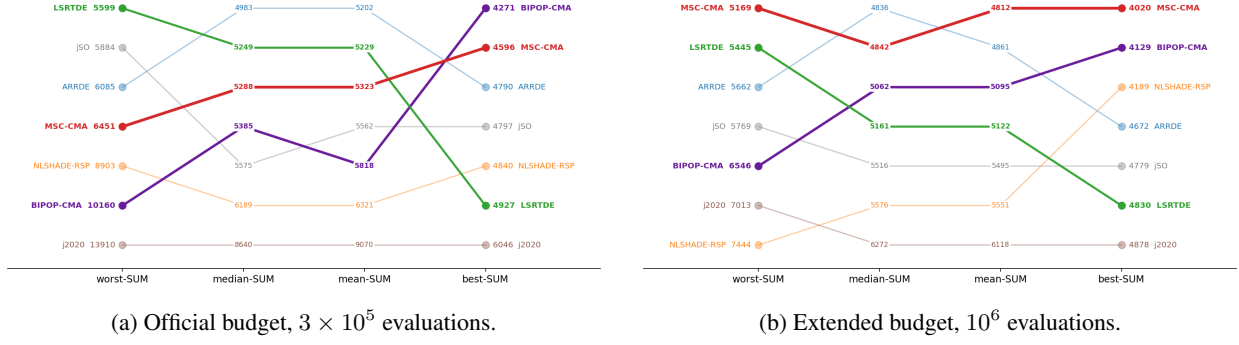


Figure 5: CEC2017 $D=30$ composition-class stress test under two evaluation budgets.

A focused $D=30$ budget-scaling diagnostic clarifies the scalability boundary of the basin-discovery mechanism. Figure 5 compares the CEC2017 $D=30$ composition class at the official CEC budget, 3×10^5 evaluations, and at an extended budget of 10^6 evaluations. This comparison is not part of the official-budget ranking; it is a diagnostic experiment outside the stated $D \leq 20$ design envelope.

At the official budget, MSC-CMA-ES remains competitive but does not dominate: it ranks fourth on worst-SUM, third on median-SUM, third on mean-SUM, and second on best-SUM. At 10^6 evaluations, however, the ordering changes: MSC-CMA-ES attains the lowest worst-SUM, mean-SUM, and best-SUM, and is nearly tied with ARRDE on median-SUM. This reversal suggests that the $D=30$ loss of dominance is not a complete disappearance of topographic signal. Rather, the benefit of structure-aware restarts is delayed: in higher dimension a fixed Sobol sample gives a sparser basin model, so more of the budget is needed before the information collected in Phase 0 is amortized by the subsequent CMA-ES restarts.

This interpretation is also consistent with exploratory landscape analysis (ELA). ELA features are estimated from finite samples, and their reliability depends on sample size, dimension, modality, and sampling strategy. Thus, sample-based topographic descriptors should be expected to become less stable when the dimension grows while the exploratory sample size remains fixed. MSC-CMA-ES inherits the same limitation: Phase 0 can guide restarts only when the nearest-better graph still resolves a useful basin signal, see [22, 18, 13].

References

- [1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proc. ACM SIGKDD*, pages 2623–2631, 2019. doi:10.1145/3292500.3330701.
- [2] A. Auger and N. Hansen. A restart CMA evolution strategy with increasing population size. In *Proc. IEEE Congress on Evolutionary Computation (CEC)*, pages 1769–1776, 2005. doi:10.1109/CEC.2005.1554902.
- [3] N. H. Awad, M. Z. Ali, J. J. Liang, B. Y. Qu, and P. N. Suganthan. Problem definitions and evaluation criteria for the CEC 2017 special session on single objective real-parameter numerical optimization. Technical report, Nanyang Technological University, 2017.
- [4] R. Biedrzycki. A version of IPOP-CMA-ES algorithm with midpoint for CEC 2017 single objective bound constrained problems. In *Proc. IEEE CEC*, pages 1489–1494, 2017. doi:10.1109/CEC.2017.7969479.
- [5] J. Brest, M. S. Maučec, and B. Bošković. Single objective real-parameter optimization: Algorithm jSO. In *Proc. IEEE CEC*, pages 1311–1318, 2017. doi:10.1109/CEC.2017.7969456.
- [6] J. Brest, M. S. Maučec, and B. Bošković. Differential evolution algorithm for single objective bound-constrained optimization: Algorithm j2020. In *Proc. IEEE CEC*, pages 1–8, 2020. doi:10.1109/CEC48606.2020.9185551.
- [7] N. Hansen and A. Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001. doi:10.1162/106365601750190398.
- [8] N. Hansen. Benchmarking a BI-population CMA-ES on the BBOB-2009 function testbed. In *Proc. GECCO Companion*, pages 2389–2396, 2009. doi:10.1145/1570256.1570333.
- [9] N. Hansen, Y. Akimoto, and P. Baudis. CMA-ES/pycma on Github. Zenodo, 2019. <https://github.com/CMA-ES/pycma>. doi:10.5281/zenodo.2559634.

- [10] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. COCO: A platform for comparing continuous optimizers in a black-box setting. *Optimization Methods and Software*, 36(1):114–144, 2021. doi:10.1080/10556788.2020.1808977.
- [11] N. Hansen, A. Auger, D. Brockhoff, and T. Tušar. Anytime performance assessment in blackbox optimization benchmarking. *IEEE Transactions on Evolutionary Computation*, 26(6):1293–1305, 2022. doi:10.1109/TEVC.2022.3210897.
- [12] J. J. Liang, B. Y. Qu, and P. N. Suganthan. Problem definitions and evaluation criteria for the CEC 2014 special session on single objective real-parameter numerical optimization. Technical report, Zhengzhou University / Nanyang Technological University, 2013.
- [13] M. A. Muñoz, M. Kirley, and K. Smith-Miles. Analyzing randomness effects on the reliability of landscape analysis. *Natural Computing*, 21:131–154, 2022. doi:10.1007/s11047-021-09847-1.
- [14] K. F. Muzakka, S. Möller, and M. Finsterbusch. Minion: a high-performance derivative-free optimization library. <https://github.com/khoirulmuzakka/Minion>, 2025.
- [15] K. F. Muzakka, A. H. Shali, H. Suhendar, S. Möller, and M. Finsterbusch. Robust differential evolution via nonlinear population size reduction and adaptive restart: The ARRDE algorithm. arXiv:2511.18429, 2025. doi:10.48550/arXiv.2511.18429.
- [16] M. Preuss. Niching the CMA-ES via nearest-better clustering. In *Proceedings of the 12th annual conference companion on Genetic and evolutionary computation*, pages 1711–1718, 2010. doi:10.1145/1830761.1830793.
- [17] M. Preuss. Improved topological niching for real-valued global optimization. In *European Conference on the Applications of Evolutionary Computation*, pages 386–395. Springer, 2012. doi:10.1007/978-3-642-29178-4_39.
- [18] Q. Renau, C. Doerr, J. Dreö, and B. Doerr. Exploratory landscape analysis is strongly sensitive to the sampling strategy. arXiv:2006.11135, 2020. doi:10.48550/arXiv.2006.11135.
- [19] V. Stanovov, S. Akhmedova, and E. Semenkin. NL-SHADE-RSP algorithm with adaptive archive and selective pressure for CEC 2021 numerical optimization. In *Proc. IEEE CEC*, pages 809–816, 2021. doi:10.1109/CEC45853.2021.9504959.
- [20] V. Stanovov and E. Semenkin. Success rate-based adaptive differential evolution L-SRTDE for CEC 2024 competition. In *Proc. IEEE CEC*, pages 1–8, 2024. doi:10.1109/CEC60901.2024.10611907.
- [21] R. Tanabe and A. S. Fukunaga. Improving the search performance of SHADE using linear population size reduction. In *Proc. IEEE CEC*, pages 1658–1665, 2014. doi:10.1109/CEC.2014.6900380.
- [22] R. Tanabe. Towards exploratory landscape analysis for large-scale optimization: a dimensionality reduction framework. In *Proc. GECCO*, pages 546–555, 2021. doi:10.1145/3449639.3459300.
- [23] A. Törn and S. Viitanen. Topographical global optimization. In *Recent Advances in Global Optimization*, pages 384–398. Princeton University Press, 1992.
- [24] H. Wang, D. Vermetten, F. Ye, C. Doerr, and T. Bäck. IOHanalyzer: Detailed performance analyses for iterative optimization heuristics. *ACM Transactions on Evolutionary Learning and Optimization*, 2(3):1–29, 2022. doi:10.1145/3510426.
- [25] C. T. Yue, K. V. Price, P. N. Suganthan, J. J. Liang, M. Z. Ali, B. Y. Qu, N. H. Awad, and P. P. Biswas. Problem definitions and evaluation criteria for the CEC 2020 special session and competition on single objective bound constrained numerical optimization. Technical report, Zhengzhou University / Nanyang Technological University, 2019.